

Thus $\hat{p}_{ij}^{T_i} = (1 - o(1))^{\frac{\varepsilon_{\max}}{\varepsilon}} p_{ij}^\varepsilon = p_{ij}^{\varepsilon'}$ where $\varepsilon' = (1 - o(1))\varepsilon_{\max}$ for $j \in D_{s,i}$. Moreover $\hat{p}_{ij}^{T_i}$ for all $j \in D_{s,i}$ decrease proportionally from p_{ij}^ε so that they also equal the corresponding $p_{ij}^{\varepsilon'}$ values by Assumption 2.

Now if we take $T_{\text{ppo}} = \max_i T_i$, then this will hold for all states e_i in the initialized cluster C_k as each row will stop updating after time T_i . Finally, since the values $\hat{p}_{ij}$ for e_i in a different cluster do not affect the above derivation, we may repeat this argument for all explored clusters to obtain the guarantee. We remark that exploring the same cluster multiple times does not affect the outcome since the clipping operation will prevent the weights from being updated from the second time onwards. $\square$

D.3 Analysis of Distillation

Lemma D.4. $(Y_0, Y_1) \sim D_{\text{dist}}$ is distributed as $Y_0 \sim \pi^\varepsilon$, $Y_1 \sim q_o^\varepsilon(\cdot|Y_0)$ where

$$\begin{aligned} q_o^\varepsilon(x_\ell|x_k) &:= \pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(X_{\bar{\tau}_{S_o}^\varepsilon}^\varepsilon = x_\ell), \quad k \neq \ell, \\ q_o^\varepsilon(x_k|x_k) &:= 1 - \sum_{\ell \neq k} q_o^\varepsilon(x_\ell|x_k). \end{aligned}$$

Proof. The Y_0 component of D_{dist} is clearly distributed according to π^ε . Consider samples such that $X_t^\varepsilon \in C_k$, in which case $Y_0^{(t)} = x_k$ and $X_t^\varepsilon \sim \pi_k^\varepsilon$. If $X_t^\varepsilon \neq x_k$, then $Y_0^{(t)} = Y_1^{(t)} = x_k$. If $X_t^\varepsilon = x_k$, the following state $Y_1^{(t)}$ is the next return of X^ε to S_o , so that

$$\mathbb{P}_{Y_0^{(t)}}(Y_1^{(t)} = x_\ell) = \mathbb{P}_{Y_0^{(t)}}(Y_1^{(t)} = x_\ell, X_t^\varepsilon = x_k) = \pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(X_{\bar{\tau}_{S_o}^\varepsilon}^\varepsilon = x_\ell)$$

holds for all $\ell \neq k$. Comparing with the definition of q_o^ε , this shows that $Y_1 \sim q_o^\varepsilon(\cdot|Y_0)$ for $(Y_0, Y_1) \sim D_{\text{dist}}$. $\square$

Proposition D.5 (Proposition 4.1 restated). Denote the return time of q_o^ε to x_k as $\bar{\tau}_{o,x_k}^\varepsilon$. For all $k, \ell \in [K]$ with $k \neq \ell$, it holds that

$$\frac{\mathbb{P}_{x_k}(\bar{\tau}_{o,x_\ell}^\varepsilon < \bar{\tau}_{o,x_k}^\varepsilon)}{q_\star^\varepsilon(C_\ell|C_k)} = 1 + \tilde{O}\left(\frac{1}{\log M}\right).$$

We remark that the asymptotic version of this result is Theorem 5.3 of [Betz and Le Roux \(2016\)](#). It is also shown that this is the best characterization of the metastable dynamics, as the transition probabilities q_o^ε themselves cannot be made to be independent of the choice of representatives S_o even in the asymptotic limit.

Proof. Applying Corollary C.7 and Proposition C.10 to the representatives x_k, x_ℓ of C_k, C_ℓ gives

$$\frac{\pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon)}{q_\star^\varepsilon(C_\ell|C_k)} = 1 + \tilde{O}\left(\frac{1}{\log M}\right).$$

Considering the numerator, conditioning the event $\{\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon\}$ on the first return of X^ε to S_o , we have that

$$\mathbb{P}_{x_k}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) = \mathbb{P}_{x_k}(X_{\bar{\tau}_{S_o}^\varepsilon}^\varepsilon = x_\ell) + \sum_{m \neq k, \ell} \mathbb{P}_{x_k}(X_{\bar{\tau}_{S_o}^\varepsilon}^\varepsilon = x_m) \mathbb{P}_{x_m}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon)$$

and multiplying both sides by $\pi_k^\varepsilon(x_k)$ gives

$$\pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) = q_o^\varepsilon(x_\ell|x_k) + \sum_{m \neq k, \ell} q_o^\varepsilon(x_m|x_k) \mathbb{P}_{x_m}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon).$$

Now note that q_o^ε is a rescaled version of X^ε reduced to S_o where only the diagonal elements $q_o^\varepsilon(x_k|x_k)$ have been increased and all other elements have been decreased proportionally. It follows that $\mathbb{P}_{x_m}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) = \mathbb{P}_{x_m}(\bar{\tau}_{o, x_\ell}^\varepsilon < \bar{\tau}_{o, x_k}^\varepsilon)$ for all $m \neq k, \ell$. Therefore applying the same argument to the chain q_o^ε we obtain

$$\begin{aligned} \pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) &= q_o^\varepsilon(x_\ell|x_k) + \sum_{m \neq k, \ell} q_o^\varepsilon(x_m|x_k) \mathbb{P}_{x_m}(\bar{\tau}_{o, x_\ell}^\varepsilon < \bar{\tau}_{o, x_k}^\varepsilon) \\ &= \mathbb{P}_{x_k}(\bar{\tau}_{o, x_\ell}^\varepsilon < \bar{\tau}_{o, x_k}^\varepsilon), \end{aligned}$$

concluding the proof. $\square$

For the convergence of the distilled model $\hat{q}_Z$, we first show the following.

Lemma D.6. *Under Assumption 5, for any $k \neq \ell$, $q_o^\varepsilon(x_\ell|x_k) = \Theta(\varepsilon/M)$ if there is a sparse edge from C_k to C_ℓ and $q_o^\varepsilon(x_\ell|x_k) = 0$ otherwise.*

Proof. If there is no sparse edge from C_k to C_ℓ , the first return of X^ε to S_o starting from x_k cannot be x_ℓ , since X^ε must first travel to a different cluster $C_n \neq C_\ell$ to escape C_k where it will inevitably hit x_n . Thus $\mathbb{P}_{x_k}(X_{\bar{\tau}_{S_o}^\varepsilon}^\varepsilon = x_\ell) = 0$ and $q_o^\varepsilon(x_\ell|x_k) = 0$.

Suppose there exists a sparse edge (x', x_ℓ) where $x' \in C_k$. Note that

$$q_o^\varepsilon(x_\ell|x_k) \leq \pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(\bar{\tau}_{x_\ell}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) = q_\star^\varepsilon(C_\ell|C_k) \left(1 + \tilde{O}\left(\frac{1}{\log M}\right)\right) = O(\varepsilon/M)$$

by Corollary C.7, Proposition C.10 and Assumption 4. Moreover letting $\Gamma_{x_k, x'}$ be a simple path with positive p^ε -probability in C_k , it holds that

$$q_o^\varepsilon(x_\ell|x_k) \geq \pi_k^\varepsilon(x_k) \mathbb{P}_{x_k}(\Gamma_{x_k, x'}) p^\varepsilon(x', x_\ell) \geq \Omega(\varepsilon/M),$$

proving the assertion. $\square$

The proof of Proposition 4.2 then follows by repeating the analysis of Theorem D.1, replacing dimension $|S|$ by K , the maximum number of outgoing edges $M + d_{\text{out}}$ by the number of sparse edges d_{out} , and the probability lower threshold $\Theta(\varepsilon)$ by $\Theta(\varepsilon/M)$. The learning rate $\eta = \Theta(K)$ is justified since $\mathbb{P}(Y_0 = x_k) = \pi^\varepsilon(C_k) = \Theta(1/K)$ by Corollary C.9. This results in a convergence rate of $O(\log KT/T)$ for the initial stage and $\exp(-\Omega(\varepsilon^2 T/M^2))$ after thresholding. The details are omitted. $\square$

We also require the following dynamical properties of q_o^ε :

Lemma D.7. *The stationary distribution π_o^ε of q_o^ε on S_o satisfies $\pi_o^\varepsilon(x_k) = \Theta(1/K)$, moreover, $\mathbb{E}_{x_k}[\tau_{o, x_\ell}^\varepsilon] = O(KM/\varepsilon)$ for all k, ℓ .*

Proof. By Proposition B.10 and Proposition 4.1, we have that

$$\frac{\pi_{\circ}^{\varepsilon}(x_k)}{\pi_{\circ}^{\varepsilon}(x_{\ell})} = \frac{\mathbb{P}_{x_{\ell}}(\bar{\tau}_{\circ, x_k}^{\varepsilon} < \bar{\tau}_{\circ, x_{\ell}}^{\varepsilon})}{\mathbb{P}_{x_k}(\bar{\tau}_{\circ, x_{\ell}}^{\varepsilon} < \bar{\tau}_{\circ, x_k}^{\varepsilon})} = \frac{q_{\star}^{\varepsilon}(C_k|C_{\ell})}{q_{\star}^{\varepsilon}(C_{\ell}|C_k)} \left(1 + \tilde{O}\left(\frac{1}{\log M}\right)\right) = \Theta(1)$$

due to Assumption 4. Then each $\pi_{\circ}^{\varepsilon}(x_k)$ must be of order $1/K$. It further follows from an application of (7) that

$$\mathbb{E}_{x_k}[\tau_{\circ, x_{\ell}}^{\varepsilon}] \leq \frac{1}{\pi_{\circ}^{\varepsilon}(x_k) \mathbb{P}_{x_k}(\bar{\tau}_{\circ, x_{\ell}}^{\varepsilon} < \bar{\tau}_{\circ, x_k}^{\varepsilon})} = \Theta\left(\frac{KM}{\varepsilon}\right).$$

□

Theorem D.8 (Theorem 4.3 restated). *For all $k \neq \ell$, $\hat{q}_{\mathbf{Z}^+}(x_{\ell}|x_k) = \Theta(1)$ if there exists a sparse edge from C_k to C_{ℓ} or 0 if not. Moreover, the hitting time $\tau_{x_{\ell}}^+$ of $x_{\ell} \in S_{\circ}$ by $\hat{q}_{\mathbf{Z}^+}$ satisfies $\mathbb{E}_{x_k}[\tau_{x_{\ell}}^+] = O(K)$.*

Proof. The chain $q_{\circ}^{\varepsilon}$ can be retrieved from $\hat{q}_{\mathbf{Z}^+}$ by making it ‘lazy.’ Indeed, note that $\hat{q}_{\mathbf{Z}^+}$ is computed from the distilled model $q_{\circ}^{\varepsilon}$ as

$$\hat{q}_{\mathbf{Z}^+}(x_{\ell}|x_k) = \begin{cases} \frac{e^{\beta} q_{\circ}^{\varepsilon}(x_{\ell}|x_k)}{q_{\circ}^{\varepsilon}(x_k|x_k) + e^{\beta} \sum_{\ell' \neq k} q_{\circ}^{\varepsilon}(x_{\ell'}|x_k)} & \ell \neq k, \\ \frac{q_{\circ}^{\varepsilon}(x_k|x_k)}{q_{\circ}^{\varepsilon}(x_k|x_k) + e^{\beta} \sum_{\ell' \neq k} q_{\circ}^{\varepsilon}(x_{\ell'}|x_k)} & \ell = k. \end{cases}$$

Since the sum in the denominator is over at most $d_{\text{out}} + 1$ nonzero terms, by choosing $e^{\beta} = \Theta(M/\varepsilon)$ we ensure that $\hat{q}_{\mathbf{Z}^+}(x_{\ell}|x_k) = \Theta(1)$ for those terms that are nonzero. Conversely, by viewing the logits of $q_{\circ}^{\varepsilon}$ as obtained by subtracting β from $\mathbf{Z}^+$, we have

$$q_{\circ}^{\varepsilon}(x_{\ell}|x_k) = \begin{cases} \frac{e^{-\beta} \hat{q}_{\mathbf{Z}^+}(x_{\ell}|x_k)}{\hat{q}_{\mathbf{Z}^+}(x_k|x_k) + e^{-\beta} \sum_{\ell' \neq k} \hat{q}_{\mathbf{Z}^+}(x_{\ell'}|x_k)} & \ell \neq k, \\ \frac{\hat{q}_{\mathbf{Z}^+}(x_k|x_k)}{\hat{q}_{\mathbf{Z}^+}(x_k|x_k) + e^{-\beta} \sum_{\ell' \neq k} \hat{q}_{\mathbf{Z}^+}(x_{\ell'}|x_k)} & \ell = k. \end{cases}$$

This may be expressed as

$$q_{\circ}^{\varepsilon}(x_{\ell}|x_k) = \lambda_k \hat{q}_{\mathbf{Z}^+}(x_{\ell}|x_k) + (1 - \lambda_k) \delta_{k\ell}$$

where

$$\lambda_k = \frac{1}{e^{\beta} \hat{q}_{\mathbf{Z}^+}(x_k|x_k) + \sum_{\ell' \neq k} \hat{q}_{\mathbf{Z}^+}(x_{\ell'}|x_k)} = e^{-\beta} q_{\circ}^{\varepsilon}(x_k|x_k) + \sum_{\ell' \neq k} q_{\circ}^{\varepsilon}(x_{\ell'}|x_k) = O(\varepsilon/M).$$

Hence $q_{\circ}^{\varepsilon}$ is equivalent to the lazy chain obtained from $\hat{q}_{\mathbf{Z}^+}$ by inserting additional self-transitions of each x_k with probability $1 - \lambda_k$. It follows that the expected hitting time $\mathbb{E}_{x_k}[\tau_{\circ, x_{\ell}}^{\varepsilon}]$ is larger than $\mathbb{E}_{x_k}[\tau_{x_{\ell}}^+]$ by at least a factor of $\min_k \lambda_k^{-1} = \Omega(M/\varepsilon)$. Comparing with Lemma D.7 proves the assertion. □